```
#Q1
#imp
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_wine
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

#loading dataset
wine = load_wine()
x = pd.DataFrame(data=wine.data, columns=wine.feature_names)
y= wine.target

print("Original dataset shape", x.shape)
print("First five observation\n", x.head)

#standardization
scaler =StandardScaler()
x_scaled= scaler.fit_transform(x)

#applying pca
pca = PCA()
x_pca =pca.fit_transform(x_scaled)

#explained variance
explained_variance = pca.explained_variance_ratio_
print("Explained variance ratio", explained_variance)

#cumulative explained variance
cumulative_variance = explained_variance.cumsum()
print("Cumulative explained variance", cumulative_variance)

#number of components explaining at least 90%
n_components = (cumulative_variance >= 0.9).argmax() +1
print("Number of components explaining at least 90% variance", n_components)
#Transform using selected components
pca_reduced = PCA(n_components=n_components)
x_reduced = pca_reduced.fit_transform(x_scaled)

#reduced dataset shape
print("Reduced dataset shape", x_reduced.shape)
```

```
Original dataset shape (178, 13)
First five observation
 <bound method NDFrame.head of      alcohol  malic_acid   ash  alcalinity_of_ash  magnesium  total_phenols  \
0      14.23        1.71  2.43               15.6      127.0           2.80
1      13.20        1.78  2.14               11.2      100.0           2.65
2      13.16        2.36  2.67               18.6      101.0           2.80
3      14.37        1.95  2.50               16.8      113.0           3.85
4      13.24        2.59  2.87               21.0      118.0           2.80
..       ...         ...   ...                ...        ...           ...
173    13.71        5.65  2.45               20.5       95.0           1.68
174    13.40        3.91  2.48               23.0      102.0           1.80
175    13.27        4.28  2.26               20.0      120.0           1.59
176    13.17        2.59  2.37               20.0      120.0           1.65
177    14.13        4.10  2.74               24.5       96.0           2.05

     flavanoids  nonflavanoid_phenols  proanthocyanins  color_intensity   hue  \
0          3.06                  0.28             2.29             5.64  1.04
1          2.76                  0.26             1.28             4.38  1.05
2          3.24                  0.30             2.81             5.68  1.03
3          3.49                  0.24             2.18             7.80  0.86
4          2.69                  0.39             1.82             4.32  1.04
..          ...                   ...              ...              ...   ...
173        0.61                  0.52             1.06             7.70  0.64
174        0.75                  0.43             1.41             7.30  0.70
175        0.69                  0.43             1.35            10.20  0.59
176        0.68                  0.53             1.46             9.30  0.60
177        0.76                  0.56             1.35             9.20  0.61

     od280/od315_of_diluted_wines  proline
0                            3.92   1065.0
1                            3.40   1050.0
2                            3.17   1185.0
3                            3.45   1480.0
```

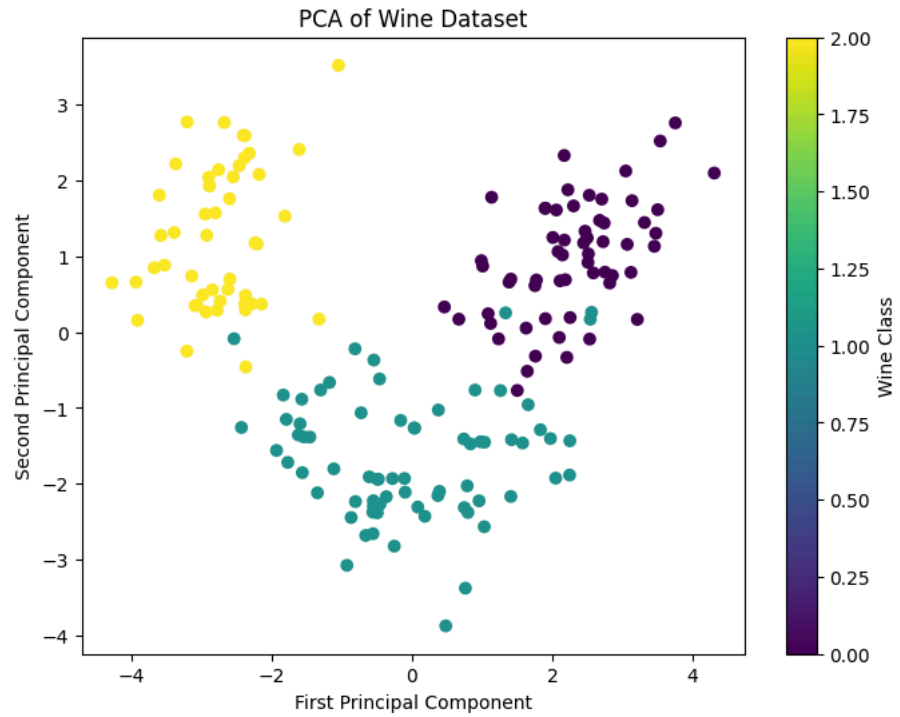
```
4                             2.93    735.0
..                             ...      ...
173                           1.74    740.0
174                           1.56    750.0
175                           1.56    835.0
176                           1.62    840.0
177                           1.60    560.0

[178 rows x 13 columns]>
Explained variance ratio [0.36198848 0.1920749  0.11123631 0.0706903  0.06563294 0.04935823
 0.04238679 0.02680749 0.02222153 0.01930019 0.01736836 0.01298233
 0.00795215]
Cumulative explained variance [0.36198848 0.55406338 0.66529969 0.73598999 0.80162293 0.85098116
 0.89336795 0.92017544 0.94239698 0.96169717 0.97906553 0.99204785
 1.        ]
Number of components explaining at least 90% variance 8
Reduced dataset shape (178, 8)
```

```
#Q2
#PCA with two components
pca_2 = PCA(n_components=2)
x_pca_2 = pca_2.fit_transform(x_scaled)
print(x_pca_2.shape)

plt.figure(figsize=(8,6))
plt.scatter(x_pca_2[:,0], x_pca_2[:,1], c=y)
plt.xlabel('First Principal Component')
plt.ylabel('Second Principal Component')
plt.title('PCA of Wine Dataset')
plt.colorbar(label="Wine Class")
plt.show()
```

```
(178, 2)
```




```
#Q3: Cancer Dataset
import numpy as np
from sklearn.datasets import load_breast_cancer
cancer = load_breast_cancer()
x = cancer.data
y = cancer.target
target_names = cancer.target_names
feature_names = cancer.feature_names

print("Target Name:\n", target_names)

print("Number of observations", x.shape[0])
print("Number of features", x.shape[1])
# Standardization
```

```
scaler = StandardScaler()
x_scaled = scaler.fit_transform(x)

# Applying PCA
pca = PCA()
x_pca = pca.fit_transform(x_scaled)

# Explained variance ratio
explained_variance = pca.explained_variance_ratio_
print("Explained variance ratio:\n", explained_variance)

# Cumulative explained variance
cumulative_variance = explained_variance.cumsum()
print("Cumulative explained variance:\n", cumulative_variance)

# Number of components explaining at least 90% variance
n_components = np.argmax(cumulative_variance >= 0.90) + 1
print("Number of components explaining at least 90% variance:",
      n_components)

# Transform using selected components
pca_reduced = PCA(n_components=n_components)
x_reduced = pca_reduced.fit_transform(x_scaled)

# Reduced dataset shape
print("Reduced dataset shape:", x_reduced.shape)

# Plot cumulative explained variance
plt.plot(
    range(1, len(cumulative_variance) + 1),
    cumulative_variance,
    marker='o'
)
plt.axhline(y=0.90, color='red', linestyle='--',
            label='90% variance')
plt.xlabel("Number of Principal Components")
plt.ylabel("Cumulative Explained Variance")
plt.title("PCA - Cancer Dataset")
plt.legend()
plt.grid(True)
plt.show()
```

```
Target Name:
 ['malignant' 'benign']
Number of observations 569
Number of features 30
Explained variance ratio:
 [4.42720256e-01 1.89711820e-01 9.39316326e-02 6.60213492e-02
 5.49576849e-02 4.02452204e-02 2.25073371e-02 1.58872380e-02
 1.38964937e-02 1.16897819e-02 9.79718988e-03 8.70537901e-03
 8.04524987e-03 5.23365745e-03 3.13783217e-03 2.66209337e-03
 1.97996793e-03 1.75395945e-03 1.64925306e-03 1.03864675e-03
 9.99096464e-04 9.14646751e-04 8.11361259e-04 6.01833567e-04
 5.16042379e-04 2.72587995e-04 2.30015463e-04 5.29779290e-05
 2.49601032e-05 4.43482743e-06]
Cumulative explained variance:
 [0.44272026 0.63243208 0.72636371 0.79238506 0.84734274 0.88758796
 0.9100953  0.92598254 0.93987903 0.95156881 0.961366   0.97007138
 0.97811663 0.98335029 0.98648812 0.98915022 0.99113018 0.99288414
 0.9945334  0.99557204 0.99657114 0.99748579 0.99829715 0.99889898
 0.99941502 0.99968761 0.99991763 0.99997061 0.99999557 1.        ]
Number of components explaining at least 90% variance: 7
Reduced dataset shape: (569, 7)
```

